

PreACT 8/9 Printable Practice Exam 14

Math • Exam 04

35 questions • 40 minutes

PART A - QUESTIONS

M4-001

What is the value of $-3 + |-8|$?

- A) -11
- B) -5
- C) 5
- D) 8
- E) 11

M4-002

What value of x satisfies $9 + 3x = -6$?

- A) -5
- B) -3
- C) -1
- D) 1
- E) 5

M4-003

Two angles are complementary. If one angle measures 37° , what is the measure of the other angle?

- A) 37°
- B) 43°
- C) 53°
- D) 127°
- E) 143°

M4-004

A liquid is at 86°C when cooling begins. Its temperature decreases at a constant rate, as shown in the table.

t (minutes)	0	2	4	6
T ($^\circ\text{C}$)	86	80	74	68

Which equation gives the temperature T after t minutes?

- A) $T = 86 - 3t$
- B) $T = 86 - 6t$
- C) $T = 86 + 3t$
- D) $T = 43 - 3t$
- E) $T = 3t$

M4-005

What is the median of the data set 5, 7, 8, 11, 12, 15?

- A) 8
- B) 9
- C) 9.5

- D) 10
- E) 11

M4-006

Which of the following is 0.00072 written in scientific notation?

- A) 7.2×10^{-5}
- B) 7.2×10^{-4}
- C) 7.2×10^4
- D) 72×10^{-4}
- E) 72×10^{-3}

M4-007

Four friends share a restaurant subtotal of \$72 equally. An 8% sales tax is added before the bill is divided. How much does each friend pay?

- A) \$18.00
- B) \$18.72
- C) \$19.08
- D) \$19.44
- E) \$23.76

M4-008

Five notebooks cost \$13.75. At the same unit price, how much do 8 notebooks cost?

- A) \$18.75
- B) \$20.00
- C) \$21.25
- D) \$22.00
- E) \$24.75

M4-009

A triangular banner has a base of 14 inches and a perpendicular height of 9 inches. What is the area of the banner?

- A) 23 square inches
- B) 46 square inches
- C) 63 square inches
- D) 126 square inches
- E) 252 square inches

M4-010

A fair spinner is divided into 12 equal sections numbered 1 through 12. What is the probability of landing on a multiple of 3?

- A) $\frac{1}{4}$
- B) $\frac{1}{3}$
- C) $\frac{5}{12}$
- D) $\frac{1}{2}$
- E) $\frac{2}{3}$

M4-011

What is the value of $2^4 + \sqrt{36}$?

- A) 10
- B) 16

- C) 20
- D) 22
- E) 28

M4-012

The first four terms of a sequence are 160, 80, 40, and 20. Each term after the first is $\frac{1}{2}$ of the preceding term. What is the sixth term?

- A) 2.5
- B) 5
- C) 10
- D) 20
- E) 40

M4-013

A phone plan costs \$18 per month plus \$0.12 for each text message sent. If a customer can spend at most \$30 for the month, what is the greatest number of text messages the customer can send?

- A) 90
- B) 96
- C) 100
- D) 120
- E) 250

M4-014

Point Q has coordinates $(-3, 4)$. It is rotated 180° about the origin. What are the coordinates of the image of Q?

- A) $(-4, 3)$
- B) $(-3, -4)$
- C) $(3, -4)$
- D) $(3, 4)$
- E) $(4, -3)$

M4-015

The table shows whether 100 students participate in a school music program.

	Participates	Does not participate	Total
Grade 8	32	18	50
Grade 9	24	26	50
Total	56	44	100

One student is selected at random from the students who do not participate. What is the probability that the selected student is in Grade 9?

- A) $\frac{9}{50}$
- B) $\frac{13}{50}$
- C) $\frac{9}{22}$
- D) $\frac{13}{22}$
- E) $\frac{11}{14}$

M4-016

A faucet fills 1.5 liters every 30 seconds at a constant rate. How many liters does the faucet fill in 8 minutes?

- A) 12 liters
- B) 16 liters
- C) 20 liters

- D) 24 liters
- E) 40 liters

M4-017

A customer buys 20 items consisting only of pens and notebooks. Each pen costs \$2, each notebook costs \$5, and the total cost is \$64. How many notebooks does the customer buy?

- A) 6
- B) 8
- C) 10
- D) 12
- E) 14

M4-018

A wildlife population begins at 600 animals and decreases by 10% each year. If the same percentage decrease continues, what will the population be after 2 years?

- A) 480
- B) 486
- C) 490
- D) 500
- E) 540

M4-019

A sphere has a radius of 3 centimeters. What is its volume?

- A) 9π cubic centimeters
- B) 12π cubic centimeters
- C) 18π cubic centimeters
- D) 27π cubic centimeters
- E) 36π cubic centimeters

M4-020

A linear model $y = 2.5x + 18$ estimates the number y of visitors, in thousands, to a park x months after a new exhibit opens. According to the model, how many visitors are predicted 12 months after the exhibit opens?

- A) 30,000 visitors
- B) 43,000 visitors
- C) 48,000 visitors
- D) 50,500 visitors
- E) 66,000 visitors

M4-021

What is the value of $(10^{-2})(10^5)$?

- A) 0.001
- B) 10
- C) 100
- D) 1,000
- E) 100,000

M4-022

The equation $x^2 - x - 20 = 0$ has two solutions. What is the negative solution?

- A) -20

- B) -5
- C) -4
- D) 4
- E) 5

M4-023

A function is defined by $f(x) = -3x + 7$. For what value of x is $f(x) = -8$?

- A) -5
- B) $-1/3$
- C) $1/3$
- D) 3
- E) 5

M4-024

A rectangular lot is represented on a scale drawing by a rectangle 8 centimeters long and 5 centimeters wide. The drawing uses a scale of 1:250. What is the actual perimeter of the lot in meters?

- A) 32.5 meters
- B) 52 meters
- C) 65 meters
- D) 130 meters
- E) 650 meters

M4-025

Machine A produces items at a constant rate, as shown in the table. Machine B's total production after x minutes is given by $B = 2.5x + 20$.

x (minutes)	0	5	10
Machine A items	12	27	42

After 20 minutes, how does Machine A's production compare with Machine B's production?

- A) Machine A produces 2 fewer items.
- B) Machine A produces 2 more items.
- C) Machine A produces 8 fewer items.
- D) Machine A produces 8 more items.
- E) The machines produce the same number of items.

M4-026

A circular pond has a radius of 6 meters. A walkway 2 meters wide is built around the outside of the pond. What is the area of the walkway?

- A) 4π square meters
- B) 16π square meters
- C) 20π square meters
- D) 28π square meters
- E) 64π square meters

M4-027

What value of x satisfies $5(2x + 1) - 3(x - 4) = 45$?

- A) 3
- B) 4
- C) 5
- D) 7

E) 8

M4-028

Which of the following is an irrational number between 6 and 7?

- A) $\sqrt{35}$
- B) 6.25
- C) $\sqrt{40}$
- D) $\sqrt{49}$
- E) $\sqrt[3]{216}$

M4-029

A rectangle in the coordinate plane has vertices $(-2, 1)$, $(5, 1)$, $(5, 6)$, and $(-2, 6)$. What is its perimeter?

- A) 12 units
- B) 19 units
- C) 24 units
- D) 35 units
- E) 48 units

M4-030

The table represents a linear function f .

x	-2	0	3	5
$f(x)$	9	5	-1	?

What is the missing value $f(5)$?

- A) -9
- B) -5
- C) -3
- D) 3
- E) 5

M4-031

In a club of 40 students, 22 participate in art, 18 participate in music, and 7 participate in both art and music. One club member is selected at random. What is the probability that the selected student participates in neither art nor music?

- A) $\frac{7}{40}$
- B) $\frac{9}{40}$
- C) $\frac{13}{40}$
- D) $\frac{29}{40}$
- E) $\frac{33}{40}$

M4-032

A rectangular water tank is 1.2 meters long, 0.5 meter wide, and 0.4 meter high. The tank is 75% full. How many liters of water are in the tank? Use 1 cubic meter = 1,000 liters.

- A) 18 liters
- B) 24 liters
- C) 180 liters
- D) 240 liters
- E) 320 liters

M4-033

A market sells 30 baskets consisting only of small and large baskets. A small basket costs \$8, a large basket costs \$14, and the total revenue is \$330. How many large baskets are sold?

- A) 10
- B) 12
- C) 15
- D) 18
- E) 20

M4-034

A rectangular prism measuring 9 centimeters by 6 centimeters by 4 centimeters is built from 1-centimeter cubes. How many of the cubes have at least one face exposed on the outside of the prism?

- A) 56
- B) 108
- C) 144
- D) 160
- E) 216

M4-035

A bag contains 40 beads. Of the beads, 30% are red and $\frac{1}{4}$ are blue. All remaining beads are green. If one bead is selected at random, what is the probability that it is green?

- A) $\frac{1}{4}$
- B) $\frac{3}{10}$
- C) $\frac{2}{5}$
- D) $\frac{9}{20}$
- E) $\frac{11}{20}$

PART B - ANSWER KEY + EXPLANATIONS

M4-001 - Correct: C) 5

Explanation: $|-8| = 8$, so $-3 + |-8| = -3 + 8 = 5$.

M4-002 - Correct: A) -5

Explanation: Subtract 9 from both sides to get $3x = -15$, and then divide by 3 to get $x = -5$.

M4-003 - Correct: C) 53°

Explanation: Complementary angles sum to 90° , so the other angle measures $90^\circ - 37^\circ = 53^\circ$.

M4-004 - Correct: A) $T = 86 - 3t$

Explanation: The temperature starts at 86°C and decreases by 6°C every 2 minutes, a rate of -3°C per minute. Therefore, $T = 86 - 3t$.

M4-005 - Correct: C) 9.5

Explanation: There are 6 ordered values, so the median is the mean of the third and fourth values: $(8 + 11) \div 2 = 9.5$.

M4-006 - Correct: B) 7.2×10^{-4}

Explanation: Moving the decimal point 4 places to the right changes 0.00072 to 7.2, so $0.00072 = 7.2 \times 10^{-4}$.

M4-007 - Correct: D) \$19.44

Explanation: With 8% tax, the total is $\$72 \times 1.08 = \77.76 . Dividing equally among 4 friends gives $\$77.76 \div 4 = \19.44 .

M4-008 - Correct: D) \$22.00

Explanation: Each notebook costs $\$13.75 \div 5 = \2.75 . Eight notebooks cost $8 \times \$2.75 = \22.00 .

M4-009 - Correct: C) 63 square inches

Explanation: The area of a triangle is $(1/2) \times \text{base} \times \text{height}$, so the area is $(1/2) \times 14 \times 9 = 63$ square inches.

M4-010 - Correct: B) $1/3$

Explanation: The multiples of 3 are 3, 6, 9, and 12, giving 4 favorable sections out of 12. The probability is $4/12 = 1/3$.

M4-011 - Correct: D) 22

Explanation: $2^4 = 16$ and $\sqrt{36} = 6$, so $2^4 + \sqrt{36} = 16 + 6 = 22$.

M4-012 - Correct: B) 5

Explanation: Continuing to multiply by $1/2$ gives the fifth term 10 and the sixth term 5.

M4-013 - Correct: C) 100

Explanation: After the \$18 monthly charge, \$12 remains. At \$0.12 per message, the customer can send $\$12 \div \$0.12 = 100$ messages.

M4-014 - Correct: C) (3, -4)

Explanation: A 180° rotation about the origin maps (x, y) to $(-x, -y)$, so $(-3, 4)$ maps to $(3, -4)$.

M4-015 - Correct: D) $13/22$

Explanation: Of the 44 students who do not participate, 26 are in Grade 9. The probability is $26/44 = 13/22$.

M4-016 - Correct: D) 24 liters

Explanation: Eight minutes is 480 seconds, which contains $480 \div 30 = 16$ intervals. The faucet fills $16 \times 1.5 = 24$ liters.

M4-017 - Correct: B) 8

Explanation: Let p be pens and n be notebooks. Then $p + n = 20$ and $2p + 5n = 64$. Substituting $p = 20 - n$ gives $40 + 3n = 64$, so $n = 8$.

M4-018 - Correct: B) 486

Explanation: A 10% annual decrease leaves 90% each year. After 2 years, the population is $600 \times 0.9^2 = 600 \times 0.81 = 486$.

M4-019 - Correct: E) 36π cubic centimeters

Explanation: A sphere's volume is $(4/3)\pi r^3$. With $r = 3$, the volume is $(4/3) \times \pi \times 3^3 = (4/3) \times 27\pi = 36\pi$ cubic centimeters.

M4-020 - Correct: C) 48,000 visitors

Explanation: Substitute $x = 12$: $y = 2.5(12) + 18 = 30 + 18 = 48$. Since y is measured in thousands, the prediction is 48,000 visitors.

M4-021 - Correct: D) 1,000

Explanation: Multiply powers with the same base by adding exponents: $(10^{-2})(10^5) = 10^3 = 1,000$.

M4-022 - Correct: C) -4

Explanation: Factoring gives $x^2 - x - 20 = (x - 5)(x + 4)$, so the solutions are 5 and -4 . The negative solution is -4 .

M4-023 - Correct: E) 5

Explanation: Set $-3x + 7 = -8$. Then $-3x = -15$, so $x = 5$.

M4-024 - Correct: C) 65 meters

Explanation: The actual dimensions are $8 \times 250 = 2,000$ centimeters and $5 \times 250 = 1,250$ centimeters, or 20 meters and 12.5 meters. The perimeter is $2(20 + 12.5) = 65$ meters.

M4-025 - Correct: B) Machine A produces 2 more items.

Explanation: Machine A increases by 15 items every 5 minutes, or 3 items per minute, and starts at 12. After 20 minutes it produces $12 + 3(20) = 72$ items. Machine B produces $2.5(20) + 20 = 70$ items, so Machine A produces 2 more.

M4-026 - Correct: D) 28π square meters

Explanation: The outer radius is $6 + 2 = 8$ meters. The walkway area is $\pi(8^2) - \pi(6^2) = 64\pi - 36\pi = 28\pi$ square meters.

M4-027 - Correct: B) 4

Explanation: Expanding gives $10x + 5 - 3x + 12 = 45$. Thus, $7x + 17 = 45$, so $7x = 28$ and $x = 4$.

M4-028 - Correct: C) $\sqrt{40}$

Explanation: Since $36 < 40 < 49$, $6 < \sqrt{40} < 7$. Because 40 is not a perfect square, $\sqrt{40}$ is irrational.

M4-029 - Correct: C) 24 units

Explanation: The horizontal side length is $5 - (-2) = 7$, and the vertical side length is $6 - 1 = 5$. The perimeter is $2(7 + 5) = 24$ units.

M4-030 - Correct: B) -5

Explanation: From $x = 0$ to $x = 3$, $f(x)$ changes from 5 to -1 , giving a rate of $-6/3 = -2$. Therefore, $f(5) = 5 - 2(5) = -5$.

M4-031 - Correct: A) $7/40$

Explanation: The number participating in art or music is $22 + 18 - 7 = 33$. Therefore, $40 - 33 = 7$ students participate in neither, giving probability $7/40$.

M4-032 - Correct: C) 180 liters

Explanation: The tank's full volume is $1.2 \times 0.5 \times 0.4 = 0.24$ cubic meter, or 240 liters. At 75% full, it contains $0.75 \times 240 = 180$ liters.

M4-033 - Correct: C) 15

Explanation: If all 30 baskets were small, the revenue would be \$240. Each large basket adds \$6, and $\$330 - \$240 = \$90$. Therefore, $\$90 \div \$6 = 15$ large baskets were sold.

M4-034 - Correct: D) 160

Explanation: The prism contains $9 \times 6 \times 4 = 216$ cubes. Cubes with no exposed face form an interior prism measuring 7 by 4 by 2, containing 56 cubes. Therefore, $216 - 56 = 160$ cubes have at least one exposed face.

M4-035 - Correct: D) $\frac{9}{20}$

Explanation: Of 40 beads, 30% are red, so there are 12 red beads, and $\frac{1}{4}$ are blue, so there are 10 blue beads. The remaining $40 - 12 - 10 = 18$ beads are green. The probability is $\frac{18}{40} = \frac{9}{20}$.